

Combining Solar Imagery and In Situ Data for Prediction of Geomagnetic Storms

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ABSTRACT

In this work, we proposed the utilization of the Most Extreme Space Weather Events (MESWE) dataset, which combines solar corona images with solar wind and interplanetary magnetic field parameters. The dataset targets periods of extreme solar activity that triggered great and severe geomagnetic storms. We developed deep learning models on the proposed dataset and compared the results quantitatively using Dynamic Time Warping. In addition, we introduced a methodology to reduce the risk of validation-test set split that can lead to misleading performance assessments. Our approach demonstrates a significantly reduced persistence effect compared to models trained with conventional methodologies in the Dst index forecasting community.

Keywords: Solar activity(1475) — Neural networks(1933) — Space weather(2037) —
Geomagnetic fields(646)

1. INTRODUCTION

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Under intense solar activity, space weather events, such as Coronal mass ejections (CMEs) can trigger geomagnetic storms (Schwenn 2006). These storms, characterized by disruptions in the magnetosphere, ionosphere, and thermosphere, are particularly impactful due to their capacity to interfere with satellite operations, communication systems, and even power grids on Earth (Oughton et al. 2017). A common parameter to measure geomagnetic disturbances is the Disturbance storm time (Dst) index (Sugiura & Wilson 1964). It represents a measure of the average intensity of the Earth’s external geomagnetic field, derived from data on north-south magnetic field fluctuations recorded by four strategically positioned ground-based magnetometers along the geomagnetic equator (Sugiura & Kamei 1991). Although some researchers argue that the Dst index alone is insufficient to fully characterize extreme space weather events (Borovsky & Shprits 2017), Dst index prediction is an extensive area of research within the Space Weather community, as it has been widely used since the late 1950s, and its observation has created a rich historical dataset. The two main approaches are Physics-based models and data-driven models. In our work, we focus on the latter-mentioned approach.

Lundstedt (1991) was one of the first to predict the Dst index using feed-forward neural networks (NN) with a one-hour lead time. Kugblenu et al. (1999) analyzed the strongest geomagnetic storms by NN. Jankovičová et al. (2002) used Principal Component Analysis to reduce input parameters for NN. Bala & Reiff (2012) proposed an improved operational forecast of Dst running in real-time. Lazzús et al. (2017) proposed NN, where weights were optimized by the Particle Swarm Optimization algorithm and a forecast model was built using only prior values of Dst. Wintoft & Wik (2018) compared Dst prediction based on data from Advanced Composition Explorer (ACE) and Deep Space Climate Observatory (DSCOVR) and reported that NN performed better when ACE data were used. The Bagging Ensemble learning method was proposed by Xu et al. (2020), combining NN, support vector regression (SVR), and long short-term memory (LSTM) network for the prediction of the Dst index. Park et al. (2021) presented an operational model to predict the Dst index combining NN and Empirical model. This model was trained on geomagnetic storms data, with the distinction between storms driven by Coronal Mass Ejection (CME) and Co-rotating Interaction Region (CIR). Hu et al.

(2022) presented the Convolutional Neural Network (CNN) trained on 51 geomagnetic storms to predict the probability of $Dst < -100$ nT one day ahead using remote-sensing data from the Solar and Heliospheric Observatory (SOHO) mission. Later, Hu et al. (2023) demonstrated a model trained on 66 geomagnetic storms, employing a Gated Recurrent Unit (GRU) to predict the Dst index up to 6 hours in advance. The model incorporated uncertainty estimation and a boosting method to improve accuracy. Telloni et al. (2023) compared seven neural network architectures and their geomagnetic prediction capabilities. Guastavino et al. (2024) analyzed input features and evaluated the prediction model's performance based on them.

We see two possible ways to engage in the current discussion on Dst prediction within this work. First, a community already knows what in-situ measurements are useful as input parameters to predict the Dst index (Lethy et al. 2018), but only a handful of works (e.g., Hu et al. 2022) incorporated remote-sensing solar corona data for Dst prediction. Second, delayed forecast in Dst index prediction are a key concern. Rapid Dst fluctuations during geomagnetic storms are difficult to predict on time, and conventional metrics like RMSE may not fully report the timeliness of the prediction. This often appears as a shift between predicted and actual values when plotted together, and it was reported in several works (Kugblenu et al. 1999; Stepanova & Pérez 2000; Wintoft & Wik 2018; Zhang et al. 2023).

Vigil, a space mission currently under development by the European Space Agency (ESA), will monitor solar activity from Lagrange Point 5 (L5) (Kraft et al. 2017). It will be equipped with instruments for in situ measurements, as well as remote sensing instruments such as an extreme ultraviolet (EUV) imager and a coronagraph to capture solar activity (Majirský et al. 2025). The viewing point for data from L5 will differ from the currently abundant observations in L1, but maximizing the utilization of Vigil data was our main motivation. This work may serve as a pathfinder study for combining imagery and in situ data for the development of data-driven prediction models. In this study, we aim to enhance the accuracy and timeliness of Dst index predictions by a combination of solar wind time series data with processed images of Solar corona from the most extreme solar activity periods in the past 30 years. Section 2 describes the dataset. Data split, computational

methods and evaluating methods are described in Section 3. Results and discussion are provided in Section 4.

2. DATA

2.1. *MESWE Dataset*

We have collected remote-sensing data and in-situ measurements for dates during the most extreme geomagnetic storms as identified in Majirský et al. (2025), using the Python pipeline provided in that work. This dataset, called MESWE (Most Extreme Space Weather Events), covers 39 events in the years 1998–2024 and includes EUV images, coronagraph images, and in-situ measurements, totaling 13 attributes, described in Table 1. Steps for dataset creation are outlined in Figure 1. Since each instrument operates at different cadences, the data have been resampled to align with the cadence of the LASCO C2 instrument. This ensures that all other measurements are paired with the timestamps of the coronagraph images. The average time interval between consecutive time steps is approximately 23 minutes. From this point onward, we will refer to each event according to its numbering in Table 5 provided in Majirský et al. (2025). The dataset excludes events 2 and 3 due to the absence of coronagraph images, which are essential for our study. Each event’s data range generally spans 10 days before and after the Dst index local minimum. On occasion, when data were missing for entire days, they had been removed, resulting in some events having a shorter time span. Events 16, 18, 20, 24, 26, and 27 were merged with preceding events, as they occurred within a 10-day range of each other. In summary we have 30 files in .csv format of extreme SWE events, where rows represent records of values at specific timestamps, while columns corresponds to attributes as described in Table 1.

2.2. *EUV Images Data*

It is well-established that Coronal Holes are correlated with solar wind parameters (Rotter et al. 2012), as they are known to be sources of high-speed streams that could impact the L1 and Earth’s geomagnetic conditions through CIRs (Chi et al. 2018). Even more, by interaction in interplanetary space with CME, they contribute to the creation of intensive geomagnetic storms (Echer et al. 2008).

Table 1. Attributes in MESWE Dataset

Attribute	Description
Q1	Pixels changed in Quadrant 1 of Coronagraph image [%]
Q2	Pixels changed in Quadrant 2 of Coronagraph image [%]
Q3	Pixels changed in Quadrant 3 of Coronagraph image [%]
Q4	Pixels changed in Quadrant 4 of Coronagraph image [%]
N_CH	Coronal holes coverage in the North region of solar disc [%]
R_CH	Coronal holes coverage in the Right region of solar disc [%]
S_CH	Coronal holes coverage in the South region of solar disc [%]
L_CH	Coronal holes coverage in the Left region of solar disc [%]
C_CH	Coronal holes coverage in the Center region of solar disc [%]
V_p	Solar wind speed [$km * s^{-1}$]
N_p	Solar wind density [cm^{-3}]
B_z	Southward component of interplanetary magnetic field [nT]
Dst	Dst index values [nT]

To capture the state of coronal holes during extreme SWE events, we used images in filter 195 Å from Extreme Ultraviolet Imaging Telescope (EIT) (Delaboudinière et al. 1995) on board the SOHO mission (Domingo et al. 1995). EIT images at 195 Å had cadency every ~ 12 minutes until August 2010. For events 33-38 that happened after this period, we used images in 193 Å filter taken by Atmospheric Imaging Assembly (AIA) (Lemen et al. 2012) on board Solar Dynamics Observatory (SDO) (Pesnell et al. 2012). To segment coronal holes in EUV images, we applied the SCSS-Net convolutional neural network introduced by Mackovjak et al. (2021). We used two separate model weights trained individually on EIT and AIA images as input data and SPoCA CH masks (Verbeeck et al. 2014) as ground-truth labels. SCSS-Net outputs are binary masks identifying coronal hole regions. We then divided the solar disc into five regions: North, Right, South, Left, and Center. For each region, we calculated the percentage of coronal hole coverage from the binary mask outputted by SCSS-Net with respect to all pixels in that region. The resulting percentages of CH pixels in each

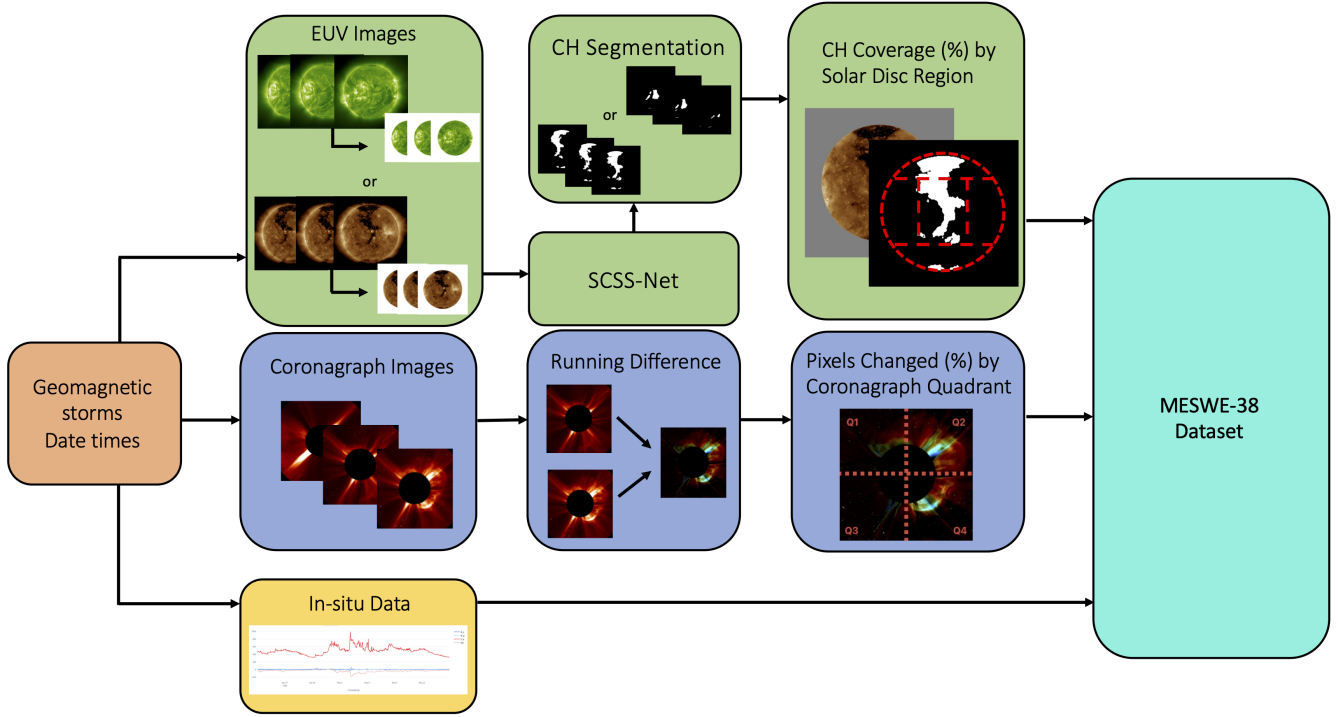


Figure 1. MESWE Dataset created by aligning remote-sensing image data with in-situ measurements for extreme SWE events. The dataset consists of 13 attributes, capturing information on coronal hole coverage across regions of the solar disc, running differences calculated in the four quadrants of coronagraph images, in-situ measurements consisting of solar wind speed, solar wind density, South-North component of magnetic field, and Dst index values.

region are five attributes of the MESWE Dataset. In this study, we refer to them as EUV visual signal features.

2.3. Coronagraph Images Data

To observe burst of CMEs during extreme SWE events, we collected coronagraph images taken by Large Angle and Spectrometric Coronagraph (LASCO) (Brueckner et al. 1995) on board SOHO. We created running difference images by subtracting consecutive coronagraph images. Pixels with

values close to zero in running difference images indicate no change between consecutive images, while non-zero pixel values suggest a sudden ejection of mass into interplanetary space, signaling a CME. To determine the direction of a CME, we divided the running difference image into four equal-sized quadrants and calculated the percentage of non-zero pixels to measure the amount of changed pixels in each quadrant. The idea behind this step is that if all four quadrants show a significant change in pixels, the CME should be directed towards Earth. This data should provide information to a machine learning model, which can be used for the timely prediction of the Dst index. The percentage of changed pixels in four equal-sized quadrants of running difference coronagraph images are four attributes of the MESWE Dataset. In this study, we refer to them as Coronagraph visual signal features. It is noted that there are more advanced techniques for identification CMEs in coronagraph images, like CACTus (Robbrecht et al. 2009), but for our approach, the presented straightforward processing is sufficient.

2.4. *In situ Data*

Solar Wind and interplanetary magnetic field (IMF) parameters measured in the L1 point are widely used for Dst index prediction. We have collected following in-situ measurements: the North-South component of the IMF from a Magnetic Field Investigation (MFI) (Lepping et al. 1995) on the WIND satellite (Harten & Clark 1995), solar wind speed, and solar wind density measured by Charge Element and Isotope Analysis System (CELIAS) (Hovestadt et al. 1995) on board SOHO. In addition, we collected hourly values of Dst (World Data Center for Geomagnetism et al. 2015). To address missing data in time-series measurements, we used the k-NN method with $k = 5$ to replace missing values with the most common values from similar occurrences. For longer gaps of missing data (i.e. 24 hours), we removed the missing data entirely, as described at the beginning of Section 2.1.

3. METHODOLOGY

Our objective is to create regression model, that will predict future values of the Dst index 10 and 20 time steps (i.e., $\sim 4-8$ hours) ahead, based on historical values of parameters from the MESWE

dataset as listed in Table 1. Our dataset is divided into three subsets: the training set for the model to learn patterns in data, the validation set for hyperparameter tuning and model selection, and the test set for reporting the model’s performance on unseen data.

3.1. *Data Split*

There is no standardized methodology for partitioning data from extreme SWE events to ensure that the validation and test sets have similar characteristics. An imbalanced split may result in a validation set that is easier for the model, leading to poor generalization to the test set, or vice versa. We wanted to deliver strong experimental results, therefore we adopted the cross-validation method (Bates et al. 2024).

While all extreme SWE events are distinct, some can be vastly different compared to others. To quantify these differences, we opted for the Wasserstein distance. Wasserstein distance calculates the minimum amount of ”work” required to transform one probability distribution into the other (Panaretos & Zemel 2019). Each event was represented as a set of empirical observations, forming a discrete probability distribution. We first computed a ground cost matrix capturing pairwise Euclidean distances between observations. The Wasserstein distance was then obtained by determining the minimal cost to transform one event’s distribution into another. These distances from one event to each other are presented in Figure 2. It can be recognized that events 19, 21, 30, 31, and partially 25 are very different to other events, particularly from events 14 and 34. To minimize selection bias and ensure that validation and test sets are as similar as possible, we created five different k-fold sets of train, validation, and test data (i.e., 5-fold cross-validation).

We observed that some events exhibit higher values in the visual signal features from coronagraph data (Q1-Q4 Attributes) before a decrease in the Dst index - this implies CME activity. To ensure a comprehensive evaluation, we categorized events as follows:

1. **Halo Event (HE):** all four attributes Q1-Q4 from Table 1 have higher values before the Dst decrease. Events: {9, 14, 15, 17, 19, 21, 30, 31, 32, 33}.

2. **Non-halo Event (NE):** not all four attributes Q1-Q4 from Table 1 show higher values before
Dst. Events: {4, 5, 6, 7, 8, 10, 22, 25, 34, 35, 36, 37, 38}.

3. **Unclassified (U):** unclear or uncertain cases. Events: {1, 11, 12, 13, 23, 28, 29}.

Example for each class is displayed in Figure 3.

Each validation and test set in every k-fold contains one event from each of these three categories, totaling three events in each set, while the remaining events are used for training. To maximize similarity between validation and test sets, we calculated an Event Sum Distance (ESD) for each event, defined as the sum of its Wasserstein distances to other events. The Set Distance (SD) is the sum of ESDs for the three events chosen for each validation or test set. We then searched all possible validation and test set combinations and selected the top five instances where the SD difference between validation and test sets was smallest. Selected k-folds are presented in Table 2.

Table 2. Selected events for each k-fold validation and test sets, ensuring sets' similarity. Remaining events were used as training data in each k-fold, so each k-fold is trained, validated, and tested on a different combination of extreme SWE events. Each validation or test set in each k-fold contains one representative event from each of the HE, NE, and U event categories.

k-fold	Validation Set Events	Test Set Events
1	4, 21, 11	8, 14, 28
2	22, 19, 28	25, 21, 1
3	4, 31, 1	34, 15, 28
4	7, 15, 13	35, 21, 1
5	7, 21, 29	35, 19, 23

3.2. Computational Methods

One question is naturally emerging. How well can the visual signal features, derived from processing coronagraph images, capture the temporal and spatial dependencies of CME propagation? Furthermore, can a machine learning model be effectively trained and leverage these dependencies?

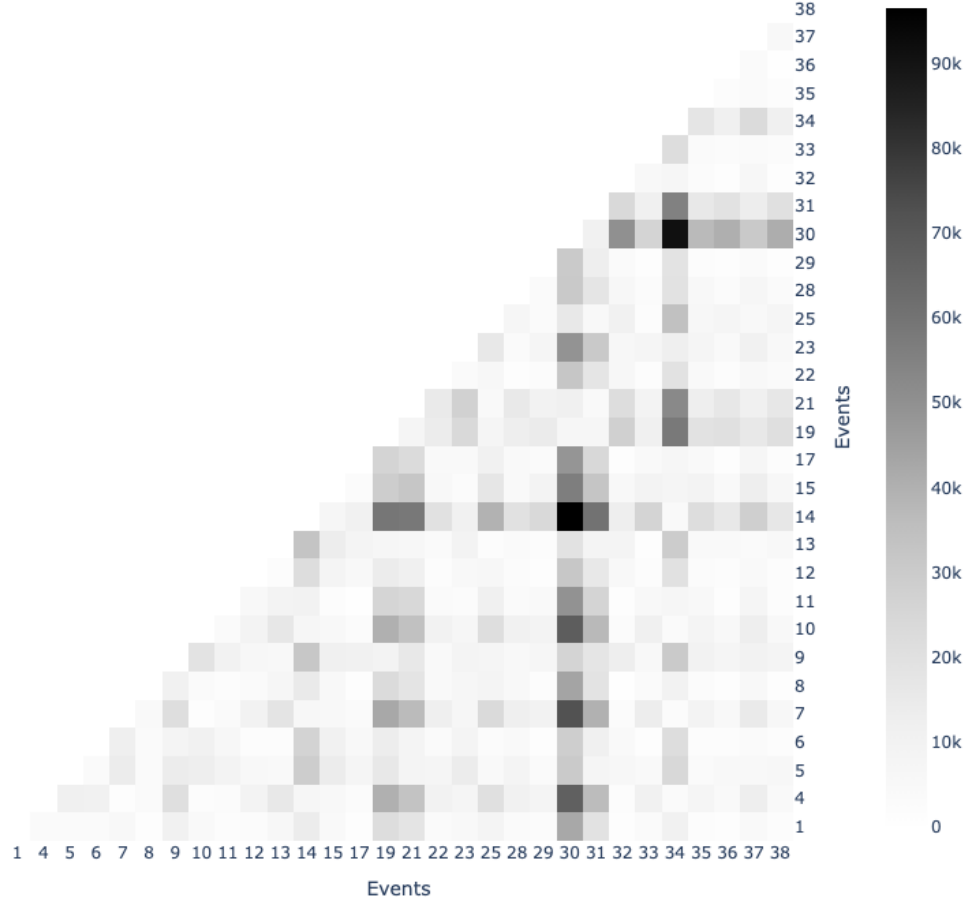


Figure 2. Wasserstein distances between one event to each other. We can observe a large distance of events 19,21,30, and 31 with respect to other extreme SWE events. For the details of the particular events, refer to Table 5 in [Majirský et al. \(2025\)](#).

To explore this, we used two approaches, *GRU-IE* and *GRU-Attn-IEC* that are described in sections below.

GRU-IE APPROACH

Gated Recurrent Units (GRUs), a type of Recurrent Neural Network (RNN), are used to model temporal dependencies in the data. Our GRU-IE approach uses In-situ data and EUV visual signal features as input. In general, RNNs are designed to process sequential data, where the current state

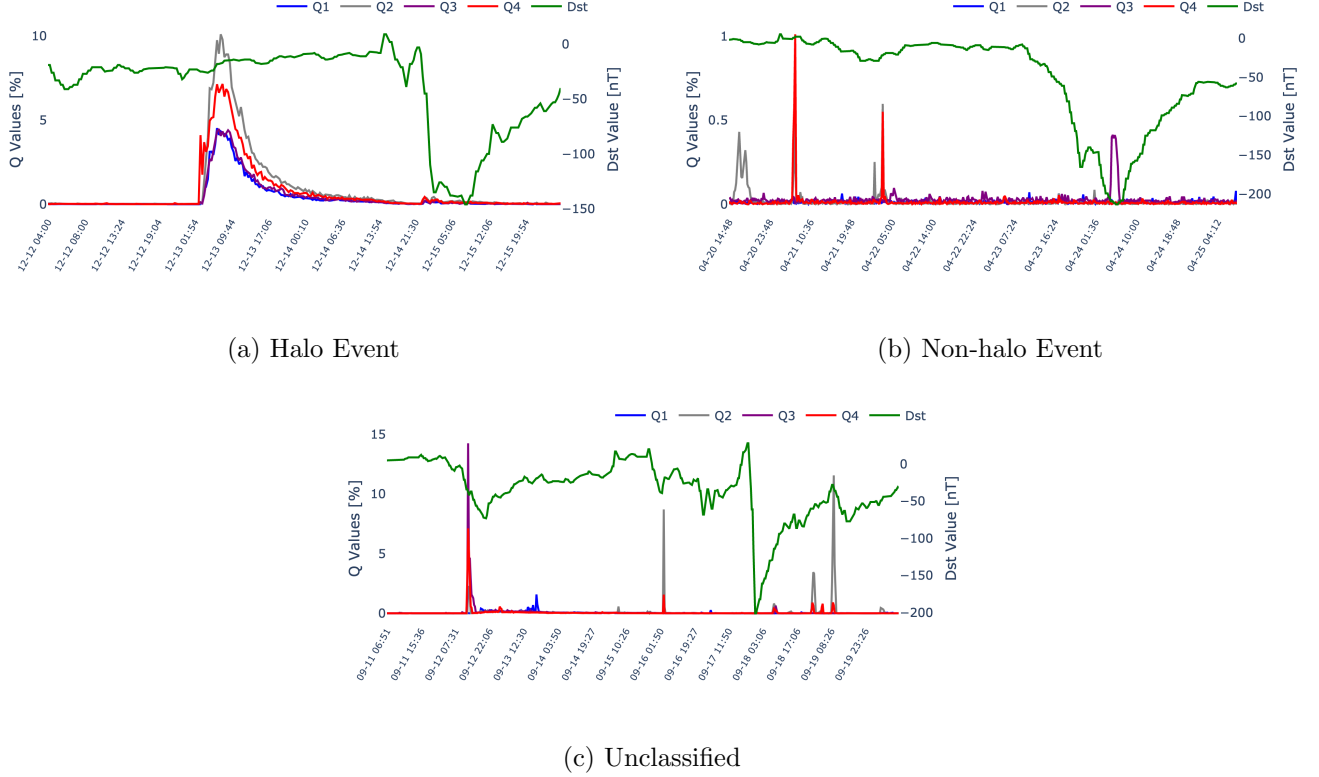


Figure 3. Different examples of categorized events. The left y-axis displays Coronagraph Visual Features values. Right y-axis displays the Dst value in nanotesla. We can observe that right before the Dst decrease, each example has a different type of activity in the Coronagraph Visual Features values. **(a)** Event 31 (2006-12-14) is categorized as Halo Event. There is visible coronagraph activity before the Dst decrease for all 4 Q-attributes with a higher percentage value. **(b)** Event 38 (2023-04-24) is categorized as Non-halo Event. Although there is visible coronagraph activity before Dst decrease, only Q4 is active and at a very low value, only around 1%. **(c)** Event 11 (2000-09-17) is categorized as Unclassified. Although there is visible coronagraph activity before the Dst decrease, not all attributes from coronagraphs are active.

of parameters depends on the previous one. However, training RNNs has been proven to be difficult due to emerging issues such as vanishing and exploding gradients (Bengio et al. 1994). Long Short-Term Memory (LSTM) (Hochreiter 1997) and Gated Recurrent Unit (GRU) (Cho 2014) are RNN architectures that address this problem. GRUs use fewer parameters than LSTMs, making them faster to train (Chung et al. 2014), which is why we used them in this work. GRU cell can be noted as follows:

$$z_t = \sigma(W_z x_t + U_z h_{t-1} + b_z) \quad (1)$$

$$r_t = \sigma(W_r x_t + U_r h_{t-1} + b_r) \quad (2)$$

$$\tilde{h}_t = \tanh(W_h x_t + U_h (h_{t-1} \odot r_t) + b_h) \quad (3)$$

$$h_t = z_t \odot \tilde{h}_t + (1 - z_t) \odot h_{t-1}, \quad (4)$$

where z_t is update gate that controls how much of the previous h_{t-1} state of parameters should be passed through a sigmoid function σ to the next state. Reset gate r_t determines how much of the previous state should be ignored in candidate hidden state $\tilde{h}_t$. In the final hidden state computation, h_t decides how much of the previous state should be retained for the current time step. W_z , W_r , W_h are parameters of inputs at time step x_t and U_z , U_r , U_h are parameters of previous state h_{t-1} . These parameters are optimized. For the initial step, the hidden state h_0 is set to zeros. After GRU, a final linear transformation projects the GRU's hidden representation to a scalar output representing the Dst index prediction. However, as predictions are processed in batches, the output takes the matrix shape [Batch, Prediction].

We conducted a systematic search over hyperparameters to identify the optimal model configuration. Best performing model has GRU with 64 hidden neurons and 2 stacked layers. To optimize model's weights effectively we used the AdamW optimizer with a learning rate 10^{-5} . To reduce overfitting, weight decay with value 0.1 along with dropout in the GRU layer set to 0.2. Empirical results revealed that applying a moving average smoothing with a window of size 5% to 10% of the input batch slightly improved model performance.

GRU-ATTN-IEC APPROACH

GRU-Attn-IEC approach uses In situ data with EUV visual signal features and also incorporates coronagraph visual signal features as an input. Coronagraph visual features are treated in a separate

branch of the model. We hypothesize that these processed signals retain the essential information present in the original coronagraph image sequences while being more computationally efficient and easier to manipulate. The first processing branch remains identical to its counterpart in the GRU-IE approach. However, the second branch, responsible for processing the coronagraph visual signal features, requires specialized attention. Self-Attention mechanism, as introduced by [Vaswani \(2017\)](#), which has become a state-of-the-art technique in the deep learning community. This mechanism allows the model to effectively capture long-range dependencies within the input sequences, enhancing the representation of temporal patterns in the coronagraph data.

The processed signals are first passed through a linear transformation followed by a non-linear activation function, Gaussian Error Linear Unit (GELU) ([Hendrycks & Gimpel 2016](#)). The transformed representations are then fed into a GRU layer, where the final hidden state is used as the query input for a multi-headed self-attention mechanism. The key and value inputs to the self-attention layer consist of all hidden states produced by the GRU. We agree that this design differs from the conventional Transformer architecture, as it retains RNN components while incorporating self-attention. However, empirical results indicate that this hybrid approach yields the best performance. While a variant eliminating the GRU altogether may be an alternative, we have not yet identified a configuration that outperforms our current model. Recent developments in state-space models, such as RWKV ([Peng et al. 2023](#)) and MAMBA ([Gu & Dao 2023](#)), offer promising approaches for reducing complexity while maintaining expressive temporal dependencies. While these methods have been predominantly applied to language modeling, their adaptation to time series tasks appears straightforward. To our knowledge, their application in space weather studies remains unexplored and might be utilized in follow-up activities.

Finally, the outputs from both processing branches are concatenated and passed through a linear transformation to obtain a scalar prediction for each batch, representing the Dst index. GRU-Attn-IEC approach is displayed in Figure 4.

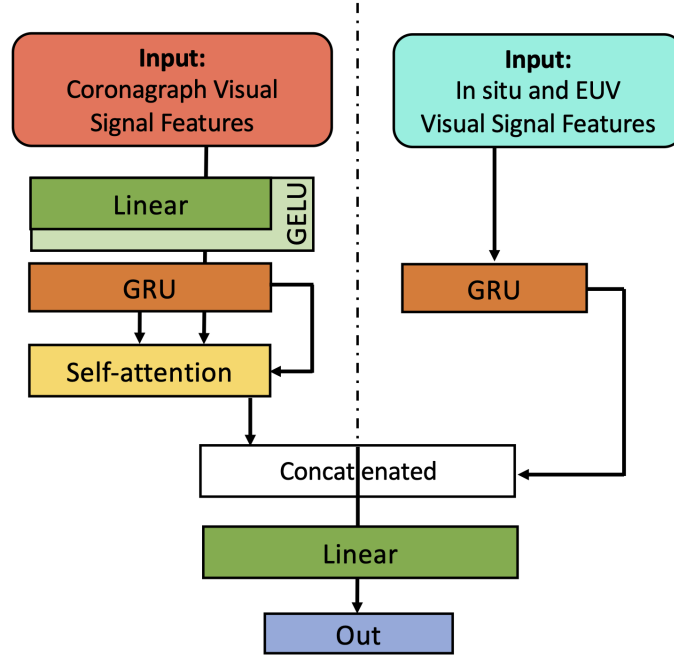


Figure 4. Deep Learning model using GRU recurrent layer with Self-attention for Coronagraph visual signal features with In situ and EUV visual signal features.

The best-performing model was configured with GRU layers containing 256 hidden neurons and 2 stacked layers in both branches. The model was optimized using the AdamW optimizer with a learning rate of 3×10^{-5} , with minimal to no regularization applied.

We experimented with a model utilizing raw coronagraph data, however, it did not provide superior results. Consequently, we do not include it in the main analysis. Nevertheless, details of this model and all our developed models are available in our codebase (SPACE::LAB et al. 2025) for those interested.

3.3. Model Evaluation

We evaluated the performance of models using 5-fold cross-validation, as described in Section 3.1. For each fold, the models were trained and tested across five distinct random seeds. Seeds were chosen once at random and were consistently applied across all models to ensure fair comparison. The performance metrics are reported as mean and standard deviation across these seeds. Root

Mean Square Error (RMSE) was employed as the primary evaluation metric, since it is commonly used in time series analysis and its adoption within studies on Dst index prediction. RMSE is given by:

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (\hat{y}_i - y_i)^2}. \quad (5)$$

Recently, [Laperre et al. \(2020\)](#) criticized the use of classical metrics such as RMSE in the evaluation of Dst index predictions, arguing that these metrics fail to capture behavior, when magnitudes of a decrease in Dst are predicted correctly but with a temporal shift, predicting the decrease after it has already occurred in the observed data. This phenomenon, referred to as persistence behavior, can be visually identified when one looks closely on predicted and actual time series data. To address this issue, authors proposed the use of Dynamic Time Warping (DTW) measure.

DTW measures similarity between time series data ([Berndt & Clifford 1994](#); [Senin 2008](#)). It seeks optimal alignment between two sequences with variations in the time axis. Given two time series $\hat{Y} = (\hat{y}_1, \hat{y}_2, \dots, \hat{y}_i)$ and $Y = (y_1, y_2, \dots, y_j)$ of length of i and j , cost matrix M of size $i \times j$ is constructed and filled as follows:

$$M(i, j) = d(\hat{y}_i, y_j) + \min\{M(i-1, j-1), M(i-1, j), M(i, j-1)\}, \quad (6)$$

where $d()$ is a distance measure, in our case, Euclidean distance. Optimal alignment is then determined using dynamic programming to find the minimum-cost warping path in the cost matrix M by backtracking. This process starts at bottom-right corner of M , $M(i, j)$ and follows the smallest neighboring value at each step until reaches the top-left corner, $M(1, 1)$. Steps taken in backtracking forms the optimal warping path which aligns $\hat{Y}$ and Y with the minimal accumulated cost. DTW can also align very distant parts of sequences. Thus, warping window w is a constraint applied in DTW, ensuring that the warping path satisfies $|i - j| \leq w$, restricting to seeking alignment only within w steps.

While our focus is on Dst index prediction, DTW has also been used to analyze solar wind time series data in Samara et al. (2022), highlighting its applicability in the space weather research community. In this study, we use the same implementation of DTW as provided in Laperre et al. (2020). From the DTW cost matrix M , one can derive valuable insights, such as: I) steps of warping path off diagonal line and II) sum of values on diagonal line. In Figure 5 are presented 3 different pairs of sequences Y and $\hat{Y}$ and corresponding DTW cost matrices. Figure 5(a) present ideal alignment, Figure 5(b) present temporal alignment, but with few magnitude differences, Figure 5(c) present temporal misalignment with magnitude changes kept same as in Figure 5(b). The situation presented in Figure 5(c) is the most common when predicting the Dst index T steps ahead. This is referred to as the persistence behavior of predictions $\hat{Y}$.

Notice how warping path changes and how values on diagonal line changes in Figures 5(d), 5(e) and 5(f). In an ideal alignment, the path follows the diagonal, and all diagonal values are zero. When time series exhibit differences in magnitude, the warping path deviates from the diagonal for a few steps, and the values along the diagonal are no longer zero. For sequences with temporal misalignment, only the first and last steps of the path remain on the diagonal line, and the values along the diagonal increase significantly. So in our model evaluation process, in addition to RMSE, we compare models using the percentage of steps on the diagonal in the cost matrix D_{on} (higher is better) and the sum of values along the diagonal line D_{sum} (lower is better).

3.4. Community Benchmark

We also compared our approach with common methodologies used in the Dst forecast community. These approaches construct long time series datasets containing solar wind and interplanetary magnetic field measurements spanning many years. This long sequence is then partitioned into training, validation, and test sets. The methodology is reported by e.g. Efitorov et al. (2018); Lethy et al. (2018); Purnomo et al. (2021); Zhang et al. (2023). We evaluated the performance of this methodology on the solar wind and IMF data from the MESWE test set, which comprises data from extreme solar activity periods. For this comparison, we obtained measurements of V_p , N_p , B_z , and Dst from 1998 to 2016 in a 1-hour cadence. Since the MESWE dataset follows the cadence of the

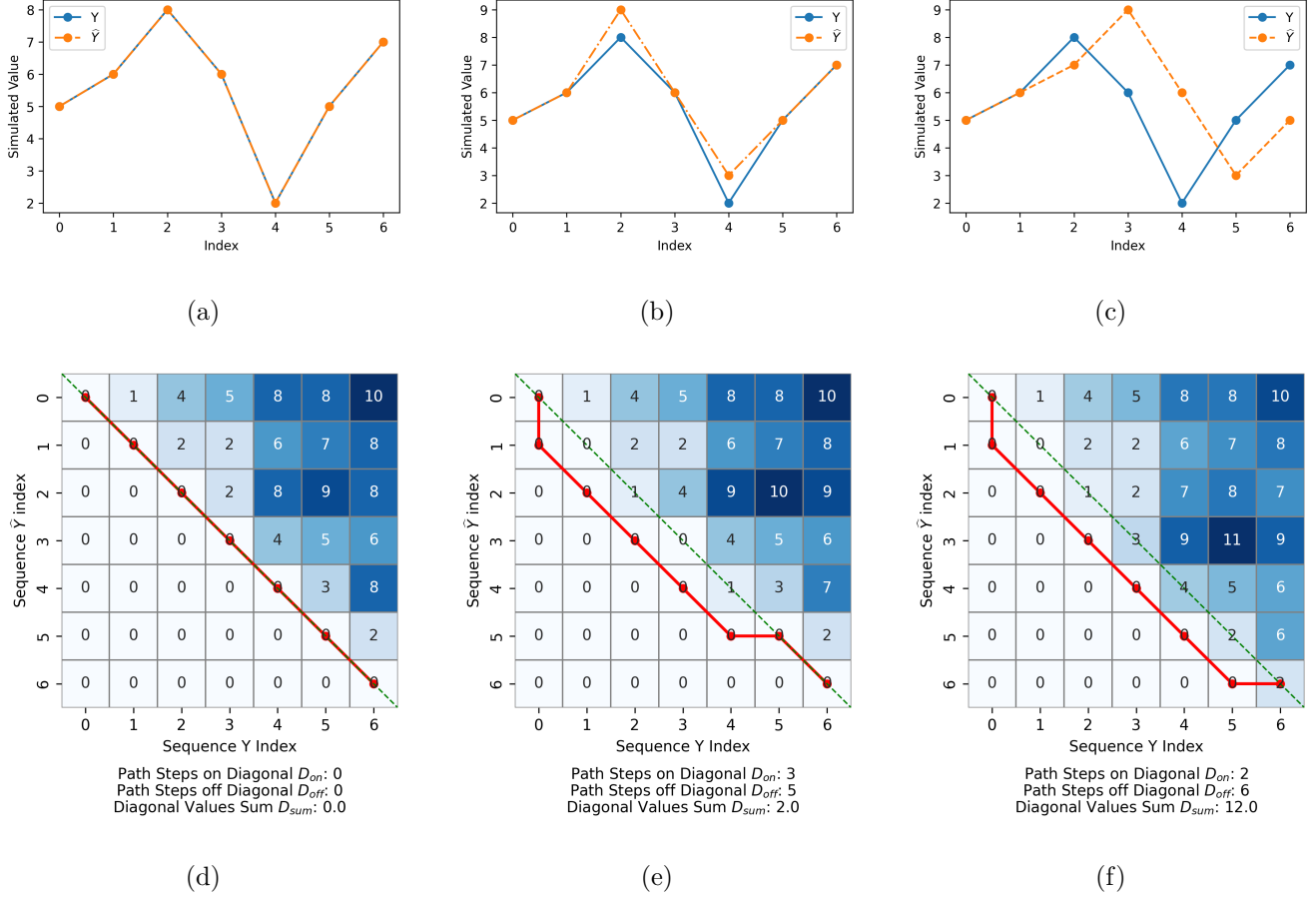


Figure 5. Demonstration of changes in the warping path and diagonal values for different time series alignments on arbitrary values. (a–c) present different simulated time series, while (d–f) depict their corresponding DTW cost matrices with warping paths. In an ideal alignment, the warping path follows the diagonal, and diagonal values remain zero. For time series with differences in magnitude, the warping path deviates slightly from the diagonal, and diagonal values are no longer zero. When sequences are misaligned in the time axis, only the first and last steps of the warping path remain on the diagonal, with a significant increase in diagonal values.

SOHO/LASCO coronagraph (20–30 min.), we interpolated the data to 30 min. cadence as well. The final 20% of time steps, period from April 2013 to December 2015, were classified as test set. This test time span includes the same extreme event from the MESWE dataset used in the test set, specifically under number 35 from the 3rd fold in Table 2. We have trained a GRU recurrent neural network and evaluated its performance on Event 35. The results were compared against those obtained using the

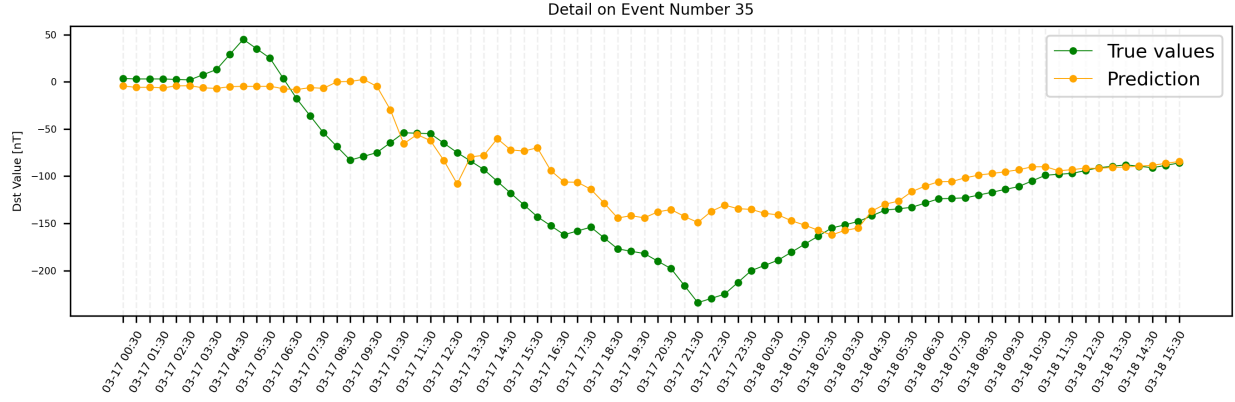
community benchmark dataset as described above, as well as the $GRU - IE$ and $GRU - Attn - IEC$ approaches on the MESWE dataset for the same event.

4. RESULTS AND DISCUSSION

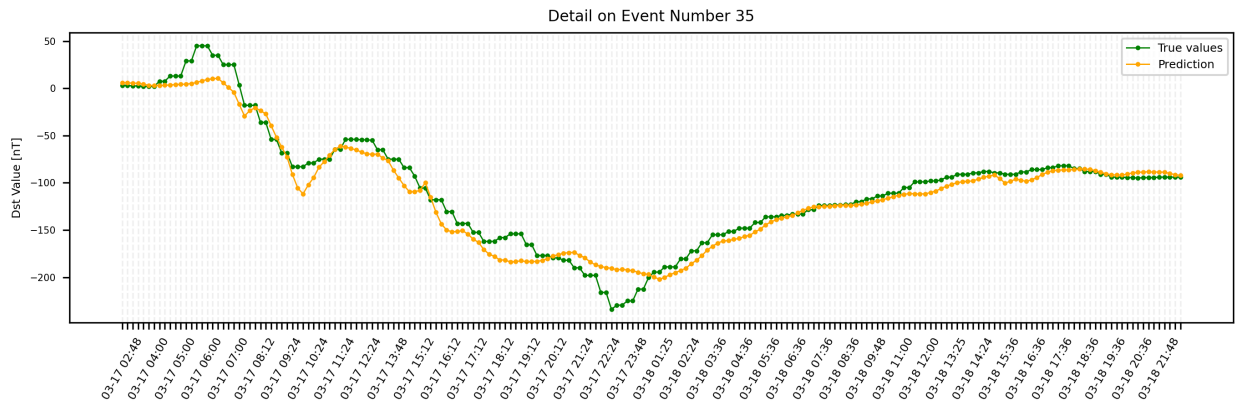
Performance of models $GRU - IE$ and $GRU - Attn - IEC$ on different k-folds as introduced in Table 2 from the MESWE dataset are presented in Table 3 for predictions 10 and 20 time steps ahead. The mean RMSE varies across k-folds for both forecasting horizons. For the 10-step prediction, $GRU - IE$ achieves a lower test RMSE of $17.0 nT$ compared to $17.4 nT$ for $GRU - Attn - IEC$. However, for the 20-step prediction, $GRU - Attn - IEC$ slightly outperforms $GRU - IE$, with RMSE values of $24.1 nT$ and $24.3 nT$, respectively.

Table 3. Experiment results for our GRU-IE and GRU-Attn-IEC approaches on test dataset expressed by evaluation metrics: $RMSE$ of Dst, D_{on} , and D_{sum} of cost DTW matrix. Results are presented for all five k-folds together with their mean values, for predictions 10 and 20 time steps ahead.

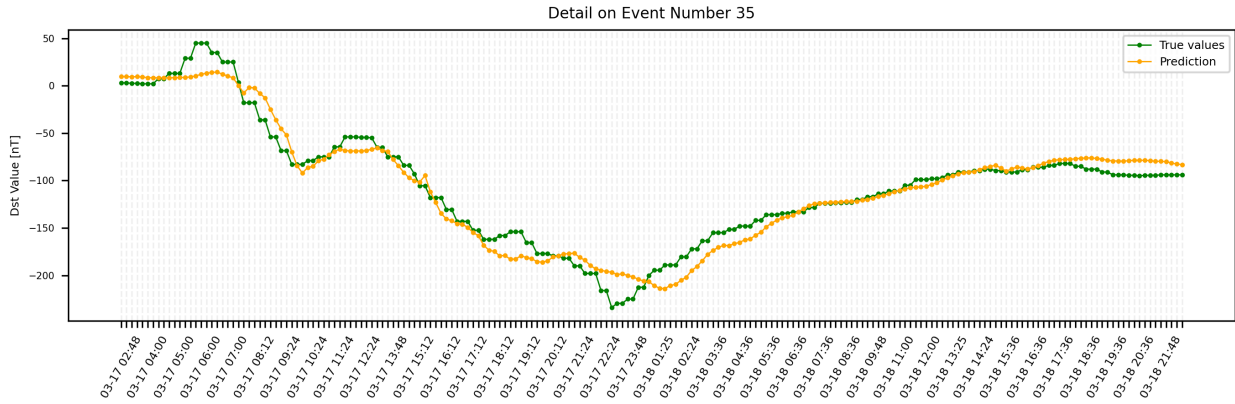
		GRU-IE			GRU-Attn-IEC		
	k-fold	RMSE [nT]	D_{on} [%]	D_{sum}	RMSE [nT]	D_{on} [%]	D_{sum}
Prediction ahead 10 steps	1	20.4±0.3	6.8±0.1	$2.15 * 10^7$	21.0±0.4	6.79±0.6	$2.22 * 10^7$
	2	17.5±0.2	5.2±0.4	$2.39 * 10^7$	18.4±0.7	4.44±0.5	$2.89 * 10^7$
	3	12.5±0.1	6.2±0.5	$2.52 * 10^7$	13.0±0.2	6.51±0.8	$2.75 * 10^7$
	4	14.2±0.3	4.9±0.5	$2.87 * 10^7$	14.6±0.2	4.98±0.4	$3.56 * 10^7$
	5	20.5±0.2	3.9±0.3	$4.58 * 10^7$	19.9±0.5	3.73±0.3	$4.65 * 10^7$
	Mean	<u>17.0</u>	<u>5.3</u>	<u>$2.9 * 10^7$</u>	<u>17.4</u>	<u>5.3</u>	<u>$3.21 * 10^7$</u>
Prediction ahead 20 steps	1	28.1±0.4	4.93±1.0	$3.46 * 10^7$	29.1±0.6	5.17±0.1	$3.15 * 10^7$
	2	24.6±0.6	2.38±0.6	$3.35 * 10^7$	24.9±0.5	2.43±0.5	$3.91 * 10^7$
	3	17.3±0.2	3.75±0.7	$3.54 * 10^7$	17.3±0.5	4.41±0.3	$3.43 * 10^7$
	4	21.6±0.3	2.48±0.1	$4.11 * 10^7$	21.5±0.4	2.47±0.1	$4.75 * 10^7$
	5	29.7±0.1	2.11±0.2	$6.66 * 10^7$	27.7±0.9	2.14±0.3	$6.45 * 10^7$
	Mean	<u>24.3</u>	<u>3.1</u>	<u>$4.22 * 10^7$</u>	<u>24.1</u>	<u>3.33</u>	<u>$4.34 * 10^7$</u>



(a) Community benchmark



(b) GRU-IE



(c) GRU-Attn-IEC

Figure 6. Example of detailed predictions ahead 10 steps for the event from the test set that occurred in March 2015 using our different approaches.

In contrast, the mean D_{sum} across k-folds is lower for $GRU - IE$ approach for both predictions horizons, meaning that it aligns predictions slightly better with observed Dst values. However, the performance of the models according to D_{on} varies with the prediction horizon length. For 10-step-ahead predictions, both models achieves same mean of D_{on} across k-folds. However, when extending the prediction horizon to 20 steps, $GRU - Attn - IEC$ model achieves a superior mean D_{on} across k-folds, exceeding $GRU - IE$ by 0.13%.

From this comparative analysis, it can be concluded that $GRU - IE$ achieves better mean values for two out of three comparative scores for 10 time steps ahead forecast, but as we extend prediction to 20 steps, it turns around into $GRU - Attn - IEC$ favor. The difference in performance between the two models remains relatively small, raising the question of whether the added complexity of $GRU - Attn - IEC$ justifies the marginal improvement in predictive accuracy.

Across different k-folds, both models demonstrate the poorest performance on the fifth fold across all comparable measures. This suggests that this fold presents greater challenges for the models in learning underlying patterns and generalizing it on the test set. This might lead to important methodological question: “How do we know if a upcoming geomagnetic storm is not from fifth fold?”.

Figure 6 presents predictions 10 steps ahead for Event 35 (March 2015) created by different approaches. Figure 6(a) demonstrates predictions based on the community benchmark dataset as described in Section 3.4, while Figure 6(b) and Figure 6(c) demonstrate predictions on MESWE dataset with $GRU - IE$ and $GRU - Attn - IEC$ approaches, respectively, as detailed in Section 3.2. Predictions in Figure 6(b) and Figure 6(c) align more closely with the target Dst values and outline a reduced persistence effect. In particular, in Figure 6(c) during Dst decline between 9:00 and 10:00 on 17 March 2015, predictions are closer to target values compared to Figure 6(b) at the same time, where Dst prediction is slightly overestimated. However, neither approach presented in Figure 6(b) nor in Figure 6(c) accurately predicted magnitude of the decrease, underestimating it by ≈ 60 nT.

Incorporating processed features from remote sensing data presents a promising direction for improving Dst index prediction, as it introduces additional information that helps mitigate the persistent

time-lag issue. While our current approach has not fully eliminated this temporal misalignment, it offers a step toward addressing it. Future work may benefit from shifting the optimization objective away from conventional metrics like RMSE and instead directly minimizing temporal discrepancies using alignment-based metrics such as Dynamic Time Warping.

5. CONCLUSION

In this work, we have proposed Most Extreme Space Weather Events (MESWE) dataset, that integrates extracted information from remote sensing images, specifically EUV and coronagraph observations with solar wind and magnetic field parameters. The dataset focuses on periods of extreme solar activity that led to great and severe geomagnetic storms. The dataset is provided in a user-friendly format, facilitating easy manipulation and immediate use by the research community.

Additionally, we demonstrated how Wasserstein distance can be utilized to partition the data, ensuring that the validation and test sets contains similar characteristics. This approach lowers the risk of models being incorrectly evaluated on the validation set while performing poorly on the test set, an issue that can emerge when data is not carefully split.

Using the MESWE dataset, we trained deep learning models incorporating GRU recurrent layers and self-attention mechanisms to predict Dst 10 and 20 steps (i.e., ~ 4 -8 hours) ahead. We demonstrated how Dynamic Time Warping (DTW) can be applied to quantitatively evaluate time series predictions and used this method to assess the performance of the proposed approaches.

The methodology and results presented in this work are directly aligned with the objectives of the ESA Vigil mission. It should be noted, however, that the L5 viewpoint will differ significantly from the current observational geometry used in our study. Moreover, there is currently no sufficiently large database of remote sensing observations from L5 that captures a large amount of extreme SWEs. Despite this, our work demonstrates how remote sensing features from EUV and coronagraph imagery, can be transformed into time-series inputs and combine with in situ time-series for machine learning models.

Data for this work were collected using the pipeline specified in our GitHub repository ([SPACE::LAB et al. 2024](#)) and the codebase for this work is presented at ([SPACE::LAB et al. 2025](#)). This work was supported by the government of Slovakia through the ESA contract No. 4000142780/23/NL/MH/mp under the RPA (Requesting Party Activity) in Slovakia. ESA Disclaimer: The view expressed herein can in no way be taken to reflect the official opinion of the European Space Agency. We are very thankful to the supervisors of this activity, namely: Nicolas Longép  (ESA/ESRIN) and Jorge Amaya (ESA/ESOC). We acknowledge our precious team member, Samuel Amrich, who passed away on 27 April 2025, for his irreplaceable contribution to the presented work.

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